01 Mouth, pharynx and salivary glands

Team: _____

Basic function (this is all you need): Food enters, is chewed, mixed with saliva, and swallowed.

STRUCTURES TO DRAW AND LABEL

- Lips, cheeks, vestibule, oral cavity proper
- Hard palate (palatine processes of the maxillae, horizontal plates of the palatine bones) and soft palate with the uvula
- Palatoglossal and palatopharyngeal arches with the palatine tonsils between them
- Tongue: body, root, lingual frenulum; papillae: filiform, fungiform, foliate, circumvallate (vallate)
- Teeth: 20 deciduous, 32 permanent; incisors, canines, premolars, molars; crown, neck, root
- Tooth layers: enamel, dentin, pulp cavity, cementum, periodontal ligament
- Parotid gland and parotid duct (opens opposite the upper second molar)
- Submandibular gland and duct (opens at the sublingual caruncle beside the frenulum)
- Sublingual gland (many small ducts along the sublingual fold)
- Pharynx: nasopharynx, oropharynx, laryngopharynx; pharyngeal constrictor muscles

NERVES AND BLOOD SUPPLY

- Parasympathetic to the parotid: CN IX (glossopharyngeal) through the otic ganglion
- Parasympathetic to the submandibular and sublingual glands: CN VII (facial) through the submandibular ganglion
- Sympathetic to all salivary glands: superior cervical ganglion
- Tongue sensation: anterior two thirds general sensation CN V3 (lingual nerve), taste CN VII (chorda tympani); posterior third both by CN IX
- Tongue muscles: CN XII (hypoglossal), except palatoglossus (CN X)
- Pharynx: pharyngeal plexus (CN IX and CN X)
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: External carotid branches: facial, lingual and maxillary arteries

THE WALL UNDER THE MICROSCOPE

- Oral lining: nonkeratinized stratified squamous epithelium. Keratinized on the gingiva and hard palate
- Tongue: skeletal muscle core running in three planes, covered by stratified squamous epithelium with papillae
- Taste buds on fungiform, foliate and circumvallate papillae (not filiform)
- Salivary glands: serous acini (dark, round nuclei, zymogen granules), mucous acini (pale, flattened basal nuclei), serous demilunes on mixed glands
- Parotid: almost all serous. Submandibular: mixed, mostly serous. Sublingual: mixed, mostly mucous
- Oropharynx and laryngopharynx: stratified squamous. Nasopharynx: pseudostratified ciliated columnar

CELLS AND WHAT THEY MAKE

Serous cells (salivary glands)	Watery fluid carrying salivary amylase and lysozyme
Mucous cells (salivary glands)	Mucin, which becomes mucus
Duct cells (striated ducts)	Adjust the salt content of saliva
Gustatory (taste) cells	Detect taste; found in taste buds
Odontoblasts	Dentin (for the life of the tooth)
Ameloblasts	Enamel, only before the tooth erupts

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: three faucets

Parotid in the cheek runs pure water (all serous). Submandibular under the jaw is mostly water with some slime (mixed, mostly serous). Sublingual under the tongue is mostly slime (mixed, mostly mucous). The lower the gland, the slimier the saliva.

Mnemonic: Every Dentist Pulls Carefully

Enamel, Dentin, Pulp, Cementum. The first three run outside to inside through the crown; cementum coats the root. Tongue taste: seven for the tip, nine for the back (CN VII anterior two thirds, CN IX posterior third).

02 Esophagus and stomach

Team: _____

Basic function (this is all you need): The esophagus moves the bolus to the stomach. The stomach churns it with acid and enzymes into chyme.

STRUCTURES TO DRAW AND LABEL

- Esophagus: about 25 cm, posterior to the trachea, passes through the esophageal hiatus of the diaphragm at T10
- Upper esophageal sphincter (skeletal) and lower esophageal sphincter (a thickening of smooth muscle, not a true anatomical sphincter)
- Stomach regions: cardia, fundus, body, pyloric part (pyloric antrum and pyloric canal), pylorus, pyloric sphincter
- Lesser curvature with the lesser omentum; greater curvature with the greater omentum
- Rugae (folds of mucosa and submucosa when the stomach is empty)
- Gastric pits leading into gastric glands

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) as the anterior and posterior vagal trunks
- Sympathetic: greater splanchnic nerves (T5 to T9) to the celiac ganglion
- Upper esophagus skeletal muscle: somatic fibers in CN X (recurrent laryngeal branches)
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Celiac trunk: left gastric, right gastric, right and left gastro-omental arteries, short gastric arteries

THE WALL UNDER THE MICROSCOPE

- Esophagus: nonkeratinized stratified squamous epithelium
- Esophageal glands (mucous) in the submucosa
- Esophageal muscularis externa: skeletal muscle in the upper third, mixed in the middle third, smooth in the lower third
- Esophagus has an adventitia, except the short abdominal part, which has a serosa
- Stomach: simple columnar epithelium made entirely of surface mucous cells; no goblet cells
- Stomach muscularis externa has three layers: inner oblique, middle circular, outer longitudinal
- Cardiac and pyloric glands are mostly mucous. Fundic (gastric) glands in the fundus and body have the full set of cells
- Serosa covers the stomach

CELLS AND WHAT THEY MAKE

Surface mucous cells	Thick alkaline mucus that protects the lining
Mucous neck cells	Thinner mucus
Parietal (oxyntic) cells	Hydrochloric acid and intrinsic factor
Chief (zymogenic) cells	Pepsinogen and gastric lipase
G cells (enteroendocrine, mostly in the pyloric antrum)	Gastrin
Other enteroendocrine cells	Local hormones such as somatostatin and histamine
Stem cells (isthmus and neck)	Replace the lining

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: a three-angle washing machine

The stomach is the only part of the tract with three muscle layers (oblique, circular, longitudinal) because it has to churn, not just push. Everywhere else has two.

Mnemonic: Parietal Pumps, Chief Chews

Parietal cells pump acid (and make intrinsic factor). Chief cells make pepsinogen, the protein chewer. G cells say Go (gastrin). Esophagus muscle: skeletal at the top, smooth at the bottom, mixed in the middle, the same order as swallowing.

03 Small intestine

Team: _____

Basic function (this is all you need): Most chemical digestion finishes here and most nutrients are absorbed here.

STRUCTURES TO DRAW AND LABEL

- Duodenum: about 25 cm, C shaped around the head of the pancreas, mostly retroperitoneal; superior, descending, inferior (horizontal) and ascending parts
- Major duodenal papilla (bile duct and main pancreatic duct enter at the hepatopancreatic ampulla) and minor duodenal papilla (accessory pancreatic duct)
- Duodenojejunal flexure held by the suspensory muscle of the duodenum (ligament of Treitz)
- Jejunum: proximal two fifths of the mesenteric small intestine, thicker wall, more circular folds
- Ileum: distal three fifths, thinner wall, Peyer's patches, ends at the ileocecal valve
- Mesentery suspends the jejunum and ileum from the posterior abdominal wall
- Surface area features: circular folds (plicae circulares), villi, microvilli (brush border)

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X)
- Sympathetic: greater and lesser splanchnic nerves to the celiac and superior mesenteric ganglia
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Superior mesenteric artery (jejunal and ileal arteries). The duodenum also gets pancreaticoduodenal branches from the gastroduodenal artery

THE WALL UNDER THE MICROSCOPE

- Simple columnar epithelium with a brush border and goblet cells
- Villi: finger shaped projections with a core of lamina propria, a capillary network and a central lacteal
- Intestinal glands (crypts of Lieberkuhn) open between the villi
- Duodenal (Brunner) glands in the submucosa: found only in the duodenum
- Peyer's patches (aggregated lymphoid nodules) in the ileum
- Muscularis externa: inner circular, outer longitudinal
- Serosa on the jejunum and ileum; adventitia on the retroperitoneal parts of the duodenum

CELLS AND WHAT THEY MAKE

Enterocytes (absorptive cells)	Brush border enzymes on the microvilli; absorb nutrients
Goblet cells	Mucus; numbers increase toward the ileum
Paneth cells (base of the crypts)	Lysozyme and defensins
Enteroendocrine cells	Cholecystokinin (CCK), secretin, and others
Duodenal gland cells	Alkaline mucus
Stem cells (crypts)	Replace the epithelium about every 3 to 6 days
M cells over Peyer's patches	Sample material and pass it to immune cells

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Mnemonic: Folds, Fingers, Fuzz

Circular folds, villi, microvilli, from largest to smallest. Each one multiplies surface area again.

Mnemonic: Brunner's in the Beginning, Peyer's at the Pole end

Duodenal (Brunner) glands mark the beginning of the small intestine; Peyer's patches mark the ileum at the far end. Paneth protects the pit: Paneth cells sit at the bottom of the crypt releasing lysozyme and defensins.

04 Accessory organs: liver, gallbladder and pancreas

Team: _____

Basic function (this is all you need): The liver makes bile, the gallbladder stores and concentrates it, and the pancreas delivers enzymes and bicarbonate. All of it empties into the duodenum.

STRUCTURES TO DRAW AND LABEL

- Liver: right hypochondriac and epigastric regions; lobes: right, left, caudate (posterior, by the IVC), quadrate (anterior, by the gallbladder)
- Falciform ligament with the round ligament (remnant of the umbilical vein) in its free edge; porta hepatis (hepatic artery proper, hepatic portal vein, hepatic ducts)
- Bile path: right and left hepatic ducts, common hepatic duct, plus cystic duct forms the bile duct, which joins the main pancreatic duct at the hepatopancreatic ampulla and opens at the major duodenal papilla (hepatopancreatic sphincter)
- Gallbladder: fundus, body, neck; cystic duct with the spiral fold
- Pancreas: retroperitoneal; head in the C of the duodenum with the uncinate process, neck, body, tail at the spleen
- Main pancreatic duct to the major papilla; accessory pancreatic duct to the minor papilla

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) to liver, gallbladder and pancreas
- Sympathetic: greater and lesser splanchnic nerves to the celiac and superior mesenteric ganglia
- Gallbladder contraction is driven mainly by CCK from the duodenum, with vagal input
- These organs are not part of the tube, so they do not have the submucosal and myenteric plexuses. Their nerves travel in plexuses along the hepatic artery and the pancreatic vessels.
- Artery: Liver: hepatic artery proper and hepatic portal vein. Gallbladder: cystic artery. Pancreas: splenic artery branches and the pancreaticoduodenal arteries

THE WALL UNDER THE MICROSCOPE

- Hepatic lobule: hexagon, central vein in the middle, plates of hepatocytes radiating out, sinusoids between the plates
- Portal triad at each corner: branch of hepatic portal vein, branch of hepatic artery, bile duct
- Bile canaliculi between hepatocytes carry bile outward, opposite to blood flow
- Gallbladder: simple columnar epithelium, folded mucosa, no submucosa, muscularis, serosa or adventitia
- Pancreas: serous acini in lobules (about 98 percent of the gland); centroacinar cells inside the acinus; no striated ducts
- Pancreatic islets: pale round clusters scattered among the acini

CELLS AND WHAT THEY MAKE

Hepatocytes	Bile (bile salts, bilirubin, cholesterol), albumin and clotting factors; store glycogen
Kupffer cells (in sinusoids)	Clear worn red cells, debris and microbes
Hepatic stellate (Ito) cells	Store vitamin A
Cholangiocytes (bile duct lining)	Add water and bicarbonate to bile
Gallbladder epithelium	Pull water and salt out of bile to concentrate it
Pancreatic acinar cells	Enzymes, most as inactive precursors: trypsinogen, chymotrypsinogen, procarboxypeptidase; plus amylase, lipase, nucleases
Centroacinar and duct cells	Bicarbonate rich fluid
Islet alpha, beta, delta, PP cells	Glucagon, insulin, somatostatin, pancreatic polypeptide

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: the lobule is a wheel with two-way traffic

Hub is the central vein, spokes are the hepatocyte plates, and the corners of the rim are the portal triads. Blood drives in from the rim to the hub. Bile drives out from the hub to the rim in the canaliculi. Opposite directions, like two lanes of a road.

Mnemonic: the bile duct adds a C each step

Hepatic ducts join to make the Common hepatic duct. Add the Cystic duct and it becomes the Common bile duct. Pancreas: the head sits in the duodenum's hug and the tail touches the spleen.

05 Large intestine, rectum and anal canal

Team: _____

Basic function (this is all you need): Water and salts are absorbed, feces are formed and stored, and defecation is controlled.

STRUCTURES TO DRAW AND LABEL

- Cecum with the ileocecal valve and the vermiform appendix
- Ascending colon (retroperitoneal), right colic (hepatic) flexure
- Transverse colon (intraperitoneal, on the transverse mesocolon), left colic (splenic) flexure
- Descending colon (retroperitoneal), sigmoid colon (on the sigmoid mesocolon)
- Rectum with its transverse rectal folds; anal canal
- Three features seen from outside: teniae coli (three longitudinal muscle bands), haustra (pouches), omental (epiploic) appendices (fat tags)
- Anal canal: anal columns, anal sinuses, pectinate line, internal anal sphincter (smooth muscle), external anal sphincter (skeletal muscle)

NERVES AND BLOOD SUPPLY

- Parasympathetic: vagus nerve (CN X) to the cecum, ascending colon and about the proximal two thirds of the transverse colon; pelvic splanchnic nerves (S2 to S4) to the rest, including the rectum
- Sympathetic: lesser and least splanchnic nerves to the superior mesenteric ganglion; lumbar splanchnic nerves to the inferior mesenteric ganglion
- External anal sphincter: pudendal nerve (somatic, S2 to S4). Internal anal sphincter: autonomic
- Enteric plexuses: submucosal (in the submucosa) and myenteric (between the circular and longitudinal muscle). Present along the whole tube.
- Artery: Superior mesenteric artery (ileocolic, right colic, middle colic) to the proximal two thirds of the transverse colon; inferior mesenteric artery (left colic, sigmoid, superior rectal) beyond that

THE WALL UNDER THE MICROSCOPE

- Simple columnar epithelium with many goblet cells; no villi; no circular folds
- Deep, straight intestinal glands (crypts) packed with goblet cells
- Abundant lymphoid nodules in the lamina propria and submucosa; the appendix is loaded with them
- Muscularis externa: complete inner circular layer; the outer longitudinal layer is gathered into the three teniae coli
- Serosa on intraperitoneal parts; adventitia on the retroperitoneal parts and the lower rectum
- Anal canal: simple columnar above the pectinate line, stratified squamous below it, keratinized at the anus

CELLS AND WHAT THEY MAKE

Absorptive cells (colonocytes)	Absorb water and electrolytes
Goblet cells	Large amounts of mucus to lubricate feces
Enteroendocrine cells	Local hormones
Stem cells (crypt base)	Replace the lining
Paneth cells	Normally absent or very few in the colon (a difference from the small intestine)

TWO MEMORY TOOLS (USE THESE OR MAKE YOUR OWN)

Analogy: a scrunchie

The teniae coli are the elastic threads of a scrunchie. They are shorter than the colon, so they gather it into pouches, the haustra. The fat tags hanging off are the omental appendices.

Mnemonic: vagus to two thirds, pelvic to the rest, pudendal to the exit

The vagus reaches to about the proximal two thirds of the transverse colon. Pelvic splanchnic nerves (S2 to S4) take over from there. The pudendal nerve controls the external anal sphincter, the voluntary one. Same split for arteries: superior mesenteric to two thirds, inferior mesenteric to the rest.